

Our residuals are this:

$$S = \sum_{p \neq q} \sum_{mn} d_{pq}^{mn} - g_p^{mm} M^{mn} \overline{g_q^{nn}}. \quad (1)$$

In this equation, we assume the gains are parallel. p and q are baselines, m and n are correlations ($\in r, l$). Repeated indices are summed. What we minimize is the norm of S , that is, $S\overline{S}$, or $|S|^2$. To this end, we compute gradient of the loss, that is, $\frac{\partial |S|^2}{\partial g_p^{mn}}$ for every m, n .

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_p^{rr}} &= \sum_q \left(-d_{pq}^{rl} g_q^{ll} \overline{m^{rl}} + \right. \\ &\quad \left. -d_{pq}^{rr} g_q^{rr} \overline{m^{rr}} + \right. \\ &\quad \left. g_p^{rr} \left(g_q^{ll} m^{rl} \overline{g_q^{ll} m^{rl}} + g_q^{rr} m^{rr} \overline{g_q^{rr} m^{rr}} \right) \right). \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_p^{ll}} &= \sum_q \left(-d_{pq}^{ll} g_q^{ll} \overline{m^{ll}} + \right. \\ &\quad \left. -d_{pq}^{lr} g_q^{rr} \overline{m^{lr}} + \right. \\ &\quad \left. g_p^{ll} \left(g_q^{ll} m^{ll} \overline{g_q^{ll} m^{ll}} + g_q^{rr} m^{lr} \overline{g_q^{rr} m^{lr}} \right) \right). \end{aligned} \quad (3)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_q^{rr}} &= \sum_p \left(-g_p^{ll} m^{lr} \overline{d_{pq}^{lr}} + \right. \\ &\quad \left. -g_p^{rr} m^{rr} \overline{d_{pq}^{rr}} + \right. \\ &\quad \left. g_q^{rr} \left(g_p^{ll} m^{lr} \overline{g_p^{ll} m^{lr}} + g_p^{rr} m^{rr} \overline{g_p^{rr} m^{rr}} \right) \right). \end{aligned} \quad (4)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_q^{ll}} &= \sum_p \left(-g_p^{ll} m^{ll} \overline{d_{pq}^{ll}} + \right. \\ &\quad \left. -g_p^{rr} m^{rl} \overline{d_{pq}^{rl}} + \right. \\ &\quad \left. g_q^{ll} \left(g_p^{ll} m^{ll} \overline{g_p^{ll} m^{ll}} + g_p^{rr} m^{rl} \overline{g_p^{rr} m^{rl}} \right) \right). \end{aligned} \quad (5)$$